MISCELLANEOUS

Job Performance Requirements

NFPA 1010

6.3.9*

6.3.14

6.3.16*

<u>**Note:**</u> **All objectives in this section are testable.**

Introduction

1. Classifications of extinguishers and fire:
    a. Class A – ordinary combustibles
    b. Class B – flammable & combustible liquids and gases
    c. Class C – energized electrical equipment
    d. Class D – combustible metals and alloys
    e. Class K – kitchen, combustible cooking oils (extremely high temperatures)
2. Extinguishing methods:
    a. Cooling – reducing the burning material below its ignition temperature
    b. Smothering – excluding oxygen from the burning process
    c. Chain breaking – interrupting the chemical chain reaction burning process
    d. Saponification – forming an oxygen excluding soapy foam surface
3. Extinguishing agents:
    a. **Water – Class A only**
    b. **AFFF (Aqueous Film Forming Foam) – Class A & B**
    c. **CO_2 (Carbon Dioxide) – Class B & C**
    d. **Clean Agent** – Class A, B, & C; replacements for Halon 1211 (Bromochlorodifluoromethane) & 1301 (Bromotrifluoromethane)
        i. FE-36 (Hexafluoropropane)
        ii. HFC (Hydrofluorocarbon)
        iii. HCCF (Hydrochlorofluorocarbon)
        iv. PFC (Perfluorocarbon)
    e. **Dry Chemical – Class A, B, & C**
        i. Sodium bicarbonate
        ii. Potassium bicarbonate
        iii. Potassium chloride
        iv. Monoammonium phosphate
    f. **Dry Powder (Metal type dependent) – Class D only**
    g. **Wet Chemical (Potassium Acetate) – Class K only**

<u>Performance Objective</u>

<u>Subject:</u> Fire Extinguishers

<u>Task:</u> Verbalize and Extinguish Incipient Fire

<u>Procedure:</u>

1. Quickly **<u>verbalize the type of fire</u>**.

   <u>Note:</u> **Class D & K fires involve extremely high temperatures and are considered IDLH situations.**

2. Confirm the proper extinguishing agent and select the extinguisher needed.
3. Prior to applying the agent onto the fire, test the extinguisher to ensure proper operation:
   a. Point nozzle in a safe direction
   b. Discharge the agent with quick, short burst
4. If outside, approach fire from the windward side.
5. Locate a path of egress.
6. To operate the extinguisher, utilize the PASS technique:
   a. P – pull the pin (Performed when testing extinguisher)
   b. A – aim the nozzle
   c. S – squeeze the handle
   d. S – sweep the agent to entirely cover the width of the fire
7. Discharge agent at the base of the fire. **Ensure NOT to plunge the agent involving Class B & K fires.**

   <u>Note:</u> Discharge AFFF allowing it to gently rain down onto the fuel surface or deflect foam off a nearby object or surface. Do NOT apply foam directly onto the fuel.

8. Although Class C extinguishing agents are nonconductive, use extreme caution when combating electrical fires. **Whenever possible, disconnect or turn off the power supply, and combat as a Class A or B fire.**
9. After fire is extinguished, back away from the fire area.
10. Tag extinguisher for recharge and detection, and/or lay extinguisher on its side.

<u>Performance Objective</u>

<u>Subject:</u> Confined Space

<u>Task:</u> Exit a Constricted Opening

<u>Note:</u> This skill is conducted under IDLH conditions. You will be in full PPE and on air throughout the entire evolution. You are alone, in a confined space, and must escape without going off air or breaking the seal of your mask. Resources available are a radio, tool, and flashlight. Dimensions of the opening are 15(w) x 18(h) inches. The width represents studs which are 16 inches on center. This makes the actual space 15(w) inches.

<u>Procedure:</u>

1. Utilize contact with wall.
2. Navigate through confined space and obstacles.
3. Attempt to pass through constricted opening.
4. Make radio transmission of **“MAYDAY”** and notify command with **“LUNARS.”**
5. **Activate PASS device to audibly alert others of your location.**
6. Use your tool to clear and remove any obstructions in the constricted opening. Sound the floor beyond the opening to ensure it is safe to pass through.
7. Loosen harness straps and doff SCBA from your back. **Utilize contact with SCBA by grasping the upper left shoulder strap with your left hand, securely holding the regulator pressure hose.**
8. Pass the SCBA through the opening. **Do NOT remove your left hand from the shoulder strap.**
9. Navigate your body through the constricted opening.
10. After clearing the opening, don the SCBA.
11. **Turn on the flashlight and direct the light up toward the ceiling to visibly alert others of your location.**

<u>Note:</u> Student is NOT exiting the building and is ONLY navigating through a constricted opening from a confined space.

<u>Note:</u> Starting at number 8 of this procedure, the skill can be performed passing your feet through the opening first while pulling and maintaining contact with the SCBA.

Performance Objective

Subject: Sprinkler Systems

Task: Verbalize Sprinkler System and Components

1. An automatic sprinkler system is an integrated system of pipes, sprinkler heads, and control valves.
2. The system is designed to activate during fires by automatically discharging enough water or extinguishing agent, to extinguish the fire or prevent its spread.
3. These systems are recognized as the most reliable of all fire protection devices. It is essential for candidates/firefighters to know and understand the types of sprinkler systems, including the components and operation of the system.
4. The following are components of a sprinkler system: Water main, control valve, FDC, one-way check valve, riser, main drain, alarm test valve, retard chamber, alarm valve, feed main, cross main, branch lines, sprinkler heads.
5. Control valves are used to stop supplying water to the system, to replace sprinklers, perform maintenance, or interrupt operations. Most main water control valves are of the indicating type and are manually operated. An indicating valve shows at a glance whether it is open or closed.
6. The following are types of indicating control valves: OS&Y (Outside Stem and Yoke) valve, PIV (Post Indicator Valve), WPIV (Wall Post Indicator Valve), PIVA (Post Indicator Valve Assembly).
7. The following are types of sprinkler systems: Wet-pipe, Dry-pipe, Deluge, Pre-action, Residential, Special Extinguishing (Wet chemical, Dry chemical, Clean agent, Carbon dioxide, Foam).